HDOC Day 44 - Solved

Question 1: Reverse each word in a sentence without changing the word order.

C Program:

```
#include <stdio.h>
#include <string.h>

void reverseWords(char s[]) {
    int i, start = 0, end, j;
    char temp;
    int n = strlen(s);
    for (i = 0; i <= n; i++) {
        if (s[i] == ' ' || s[i] == '\0') {
            end = i - 1;
            for (j = start; j < end; j++, end--) {
                temp = s[j];
                s[j] = s[end];
                s[end] = temp;
            }
            start = i + 1;
        }
    }
}

int main() {
    char s[200];
    printf("Enter a sentence: ");
    fgets(s, sizeof(s), stdin);
    s[strcspn(s, "\n")] = '\0';
    reverseWords(s);
    printf("Output: %s\n", s);
    return 0;
}
```

Sample Input and Output:

```
Enter a sentence: What is special?
Output: tahW si ?laiceps
```

Question 2: Print the initials of a name.

C Program:

```
#include <stdio.h>
#include <ctype.h>
#include <string.h>

void printInitials(char name[]) {
    int i, newWord = 1;
    for (i = 0; name[i] != '\0'; i++) {
        if (name[i] == ' ') {
            newWord = 1;
        } else if (newWord) {
            printf("%c.", toupper((unsigned char)name[i]));
            newWord = 0;
        }
    }
```

```
    printf("\n");
}

int main() {
    char name[100];
    printf("Enter a name: ");
    fgets(name, sizeof(name), stdin);
    printInitials(name);
    return 0;
}
```

Sample Input and Output:

```
Enter a name: Alexander Grahambell
Output: A.G.
```